

Varun Hiremath

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Summary

Highly accomplished Senior Staff Software Engineer and Aerospace Engineering Ph.D. with over 13 years of expertise leveraging advanced computational fluid dynamics (CFD), numerical algorithms, and high-performance computing. Proven track record designing and optimizing complex, production-grade simulation software, with specialized mastery in parallel processing, hybrid CPU/GPU architectures, and scalable combustion modeling.

Education

Cornell University

PH.D. & M.S. IN AEROSPACE ENGINEERING

- Advisor: Professor Stephen B. Pope
- GPA: 4.26/4.0
- Minor: Computational Science and Engineering

Ithaca, NY, USA

Aug 2008 - Dec 2012

Indian Institute of Technology (IIT) Madras

DUAL DEGREE B.TECH. AND M.TECH. IN AEROSPACE ENGINEERING

- Cumulative GPA: 9.32/10
- Major (Aerospace Engineering) GPA: 9.52/10
- Minor (Theoretical Computer Science) GPA: 10/10

Chennai, TN, India

July 2003 - July 2008

Work Experience

GE Aerospace

SENIOR STAFF SOFTWARE ENGINEER

- Developing a High-Order CFD solver designed to work on hybrid CPU/GPU architectures.
- Driving the overall design, development, testing, validation, and documentation efforts.
- Focused on effective simulation and design of GE engine components.

Sammamish, WA, USA

Oct 2022 - Present

Siemens PLM

ADVANCED SOFTWARE ENGINEER

- Worked on the development of CFD simulation software, *Simcenter STAR-CCM+*, written in C++ and Java.
- Collaborated and worked with different teams focused on developing the core solver framework, physics models, and GUI components.
- Primarily worked in the reacting flow team on researching, implementing, and improving the performance of various combustion models.
- Implemented Acoustic Modal Analysis post-processing tool in STAR-CCM+.
- Worked on porting STAR-CCM+ to use hybrid CPU/GPU computing.
- Gained experience with integrating and using various open-source scientific tools and libraries.

Lebanon, NH, USA

Jan 2013 - Oct 2022

Cornell University

RESEARCH & TEACHING ASSISTANT

- Ph.D. thesis on *Computationally-Efficient Implementation of Combustion Chemistry*.
- Research focused on developing computationally-efficient and scalable algorithms for performing turbulent combustion simulations.
- Developed highly efficient and scalable parallel algorithms (tested on up to 9,000 cores) for running parallel simulations.
- Developed a combined dimension-reduction and tabulation algorithm for efficient implementation of chemistry.
- Some of the developed algorithms have been incorporated into commercial CFD packages such as *STAR-CCM+* and *ANSYS Fluent*.
- Worked as a *Teaching Assistant* for the following courses: Intermediate Fluid Dynamics, Dynamics of Flight Vehicles, Fluids Laboratory.

Ithaca, NY, USA

Aug 2008 - Dec 2012

Debian GNU/Linux

DEBIAN DEVELOPER

- Developer for the *Debian GNU/Linux* project, a worldwide volunteer organization dedicated to producing a high-quality, free, Linux-based operating system. Maintaining several packages including various scientific computing, plotting and visualization packages.
- Experience with packaging, source code management, collaborative code development, version control systems, bug tracking systems, Linux operating system.

India & USA

Nov 2006 - Present

Honors & Awards

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| 2020 | Siemens Excellent Performer Award , Excellent performance in Q2 FY2020 | Siemens PLM |
| 2012 | Jayesh Prize , Outstanding Presentation in the Cornell Fluid Dynamics Seminar | Cornell University |
| 2008 | Graduate Research Fellowship , Sibley School of Mechanical and Aerospace Engineering | Cornell University |
| 2008 | Hindustan Aeronautics Limited Prize , Securing Highest GPA, Aerospace Engineering | IIT Madras, India |
| 2007 | Institute Merit Prize , Best Academic Record All Years 2004 to 2007, Aerospace Engineering | IIT Madras, India |

Primary Languages	C++, Python
Secondary Languages	Fortran, C, Java, Shell Scripting, HTML, CSS
Scientific Packages	Sundials, SuiteSparse, PETSc, SLEPc, ISAT, Chemkin
CFD Software	STAR-CCM+, Ansys Fluent, Space Claim, Enight, Pointwise
Tools	Git, SVN, Shell, CMake, Make, Matlab, JIRA, Confluence

Publications

Journal Articles

- V. Hiremath and J.E. Roman “Acoustic modal analysis with heat release fluctuations using nonlinear eigensolvers”, **Applied Mathematics and Computation**, 458, (2023)
- V. Hiremath and S. B. Pope “A Study of the Rate-Controlled Constrained-Equilibrium Dimension Reduction Method and its Different Implementations”, **Combustion Theory and Modelling**, 17(2) (2013) 260–293
- V. Hiremath, S.R. Lantz, H. Wang, S.B. Pope “Large-Scale Parallel Simulations of Turbulent Combustion using Combined Dimension Reduction and Tabulation of Chemistry”, **Proceedings of the Combustion Institute**, 34(1) (2013) 205–215
- Z. Ren, G.M. Goldin, V. Hiremath, S.B. Pope “Simulations of a turbulent non-premixed flame using combined dimension reduction and tabulation for combustion chemistry”, **Fuel**, 105 (2013) 636–644
- V. Hiremath, S.R. Lantz, H. Wang, S.B. Pope “Computationally-Efficient and Scalable Parallel Implementation of Chemistry in Simulations of Turbulent Combustion”, **Combustion and Flame**, 159(12) (2012) 3096–3109
- Z. Ren, G.M. Goldin, V. Hiremath, S.B. Pope “Reduced Description of Reactive Flows with Tabulation of Chemistry”, **Combustion Theory and Modelling**, 15(6) (2011) 827–848
- V. Hiremath, Z. Ren, S.B. Pope “Combined Dimension Reduction and Tabulation Strategy using ISAT-RCCE-GALI for the Efficient Implementation of Combustion Chemistry”, **Combustion and Flame**, 158(11) (2011) 2113–2127
- V. Hiremath, Z. Ren, S.B. Pope “A Greedy Algorithm for Species Selection in Dimension Reduction of Combustion Chemistry”, **Combustion Theory and Modelling**, 14(5) (2010) 619–652

Conference Abstracts, Presentations and Posters

- P. Pepiot, Y. Liang, V. Hiremath, S. B. Pope “A Combined Adaptive Chemistry and Tabulation Strategy for the Efficient Implementation of Detailed Combustion Chemistry”, *4th International Workshop on Model Reduction in Reacting Flows, June 19–21, 2013, San Francisco, California, USA*
- V. Hiremath, S.R. Lantz, H. Wang, S.B. Pope “Computationally-Efficient Implementation of Chemistry using Combined Dimension Reduction and Tabulation in Simulations of Turbulent Combustion”, *34th International Symposium on Combustion July 29–August 3, 2012, Warsaw, Poland*
- V. Hiremath and S. B. Pope “Combined Reduction-Tabulation based Strategies for Computationally-Efficient Implementation of Combustion Chemistry”, *Eleventh International Workshop on Measurement and Computation of Turbulent Flames, July 26–28, 2012, Darmstadt, Germany*
- V. Hiremath, S.R. Lantz, H. Wang, S.B. Pope “Computationally-Efficient Parallel Implementation of Combustion Chemistry in LES/PDF Computations”, *Technical Meeting of the Eastern States Section of the Combustion Institute (ESSCI), October 9–12, 2011, University of Connecticut, Storrs, Connecticut, USA*
- V. Hiremath, Z. Ren, S.B. Pope “Computationally-Efficient Implementation of Combustion Chemistry using ISAT-RCCE-GALI”, *7th US National Combustion Meeting, March 20–23, 2011, Georgia Institute of Technology, Atlanta, USA*
- V. Hiremath, Z. Ren, and S. B. Pope “Dimension Reduction of Combustion Chemistry using ICE-PIC and ISAT”, *Symposium on Turbulence and Combustion, August 3–4, 2009, Cornell University, Ithaca, New York, USA*
- Z. Ren, V. Hiremath and S.B. Pope “Dimension reduction and tabulation of combustion chemistry using ICE-PIC and ISAT”, *6th US National Combustion Meeting, May 17–20, 2009, Ann Arbor, Michigan, USA*
- S. B. Pope, Z. Ren, V. Hiremath (2009) “Dimension Reduction and Tabulation of Combustion Chemistry using ICE-PIC and ISAT”, *2nd International Workshop on Model Reduction in Reacting Flows, March 30–April 1, 2009, University of Notre Dame, Notre Dame, Indiana, USA*